0 Loss of significance

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Goals.

- To understand the floating point representation of real numbers.
- To avoid loss of significance.

Let's talk on the first topic.

1 Floating point representation of real numbers

Definition 1.1. (floating point form.) Every real number (except zero) can be written as

$$\pm 0.m_1m_2m_3\cdots \times 10^n, \quad n \text{ is an integer}, m_1 \neq 0, \tag{1}$$

n is called the exponent, and $m_1m_2\cdots$ is called the mantissa.

For instance,

Example 1.2.

$$\frac{1}{3} = 0.3333\cdots \times 10^0, \tag{2}$$

$$1 = 0.1 \times 10^1, \tag{3}$$

The irrational number,

$$\pi = 3.14159265\cdots = 0.314159265\cdots \times 10^1 \tag{4}$$

A question arises:

Question 1. How to represent or store a number like $\frac{1}{3}, 1, \pi$ on K-decimal digit computer? For instance, we assume $K = 3$.

In order to store a number, here are steps:

1. Check if the number is 0, (0 is stored in a specific way.)
2. Look at the sign.
3. Write the unsigned number in floating point form

$$0.m_1m_2m_3\cdots \times 10^n, n \text{ is an integer, } m_1 \neq 0. \tag{5}$$

4. Limit magnitude ($n_{\min}$ is negative and $n_{\max}$ is positive.)

$$n_{\min} \leq n \leq n_{\max}. \tag{6}$$

For example, in MATLAB, the magnitudes of the double precision numbers are more or less limited like

$$-323 \leq n \leq 308. \tag{7}$$

- If $n > n_{\max}$, assign ∞.
- If $n < n_{\min}$, assign 0.

```
>> 2^1023
ans =
     8.988465674311580e+307
>> 2^1024
ans =
     Inf
>> 2^(-1074)
ans =
     4.940656458412465e-324
>> 2^(-1075)
ans =
     0
>> eps % MACHINE EPSILON
ans =
     2.220446049250313e-016
```

5. Limit precision: store a fixed number of digits, say K, (round-off the last one!).

   For example, in MATLAB, the double precision is of 16 digits.

   Round-off!

$$0.m_1m_2\cdots \approx 0.m_1m_2\cdots m_{K-1}\widetilde{m}_K, \tag{8}$$

   where

$$\widetilde{m}_K = \begin{cases} m_K, & \text{if } m_{K+1} \in \{0,1,2,3,4\} \\ 1+m_K, & \text{if } m_{K+1} \in \{5,6,7,8,9\} \end{cases} \tag{9}$$

Answer to Question 1. "Assuming there is a 3-decimal digits computer. Write $\frac{1}{3}, 1, \pi$ in 3-decimal digit computer. "

$$fl\left(\frac{1}{3}\right) = 0.333 \times 10^0, \tag{10}$$

$$fl(1) = 0.100 \times 10^1, \tag{11}$$

$$fl(\pi) = 0.314 \times 10^1. \tag{12}$$

2 Loss of significance

The number of significant digits in a decimal number starts from the first nonzero to the last digit being presented.

For example, the number of significant digits for 0.015 is 2. The number of significant digits for 0.156 is 3.

The loss of significance could typically happen when "one loses significant digits in subtraction of two numbers that are very close to each other".

- We now demonstrate this phenomena through a simple example.

$$\begin{array}{rr} & 123.4567 \\ - & 123.4566 \\ \hline & 000.0001 \end{array}$$

Through this subtraction, we lost 6 significant digits.

Let's take an example to show the loss of significant digits is bad.

Example 2.1. Calculate $\sqrt{9.01} - 3$ on a three-decimal-digit computer.

Since $\sqrt{9.01} \approx 3.0016662$, when we store this intermediate result to three significant digits we get 3.00. Subtracting 3.00, we get a final answer of 0.00. No significant digits in our answer are correct. In fact,

$$\sqrt{9.01} - 3 \approx 1.6662 \times 10^{-3}. \tag{13}$$

To avoid this problem, we use a trick called "rationalizing the numerator."

$$\begin{aligned} \sqrt{9.01} - 3 &= \frac{(\sqrt{9.01} - 3)(\sqrt{9.01} + 3)}{\sqrt{9.01} + 3} \\ &= \frac{9.01 - 3^2}{\sqrt{9.01} + 3} \\ &= \frac{0.01}{3.00 + 3} = \frac{0.01}{6} = 0.00167 \approx 1.67 \times 10^{-3}. \end{aligned}$$

Now we know a trick "rationalizing the numerator" to restructure the calculation, in order to avoid the subtraction by two numbers very close to each other.

Often, specific identities can be used, as with trigonometric expressions like

$$\sin^2 x + \cos^2 x = 1, \cdots . \tag{14}$$

Example 2.2. We have $E_1 = \frac{1-\cos x}{\sin^2 x}$. Use the double precision of MATLAB computations, we get the following table:

1. As you can see, calculation of $1 - \cos x$ when x is close to zero (or $2k\pi$, where k is an integer,) is subject to loss of significance.
2. We do a "rationalizing the numerator" by multiplying the numerator and denominator of E_1 by $1+\cos x$

$$E_1 = \frac{1 - \cos x}{\sin^2 x} \frac{1 + \cos x}{1 + \cos x} = \frac{1 - \cos^2 x}{\sin^2 x (1 + \cos x)} = \frac{1}{1 + \cos x} = E_2. \tag{15}$$

Example 2.3. Find both roots of the quadratic equation $x^2 + 9^{12}x = 3$, with 4 digits. The roots of a quadratic equation of form $ax^2 + bx + c = 0$ are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{16}$$

For our problem, this translates to

$$x = \frac{-9^{12} \pm \sqrt{9^{24} + 4(3)}}{2} \tag{17}$$

x	E_1	E_2
1.00000000000000	0.64922320520476	0.64922320520476
0.10000000000000	0.50125208628858	0.50125208628857
0.01000000000000	0.50001250020848	0.50001250020834
0.00100000000000	0.50000012499219	0.50000012500002
0.00010000000000	0.49999999862793	0.50000000125000
0.00001000000000	0.50000004138685	0.50000000001250
0.00000100000000	0.50004445029134	0.50000000000013
0.00000010000000	0.49960036108132	0.50000000000000
0.00000001000000	0.00000000000000	0.50000000000000
0.00000000100000	0.00000000000000	0.50000000000000
0.00000000010000	0.00000000000000	0.50000000000000
0.00000000001000	0.00000000000000	0.50000000000000
0.00000000000100	0.00000000000000	0.50000000000000

- Using the minus sign gives the root
$$x_1 = -2.824 \times 10^{11}. \tag{18}$$
correct to four significant digits.

- For the plus sign root
$$x_2 = \frac{-9^{12} + \sqrt{9^{24} + 4(3)}}{2}, \tag{19}$$
Matlab calculates 0.

★ Because we use MATLAB in double precision, MATLAB computes with approximately 16 significant digit. So $9^{12} = \sqrt{9^{24} + 4(3)}$. Can this calculation be saved? We must fix the loss of significance problem. The correct way to compute x_2 is by restructuring the quadratic formula:

$$\begin{aligned} x_2 &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{(-b + \sqrt{b^2 - 4ac})(b + \sqrt{b^2 - 4ac})}{2a(b + \sqrt{b^2 - 4ac})} \\ &= \frac{-4ac}{2a(b + \sqrt{b^2 - 4ac})} \\ &= \frac{-2c}{b + \sqrt{b^2 - 4ac}}. \end{aligned}$$

Substituting a, b, c for our example yields, according to MATLAB, $x_2 = 1.062 \times 10^{-11}$, which is correct to four significant digits of accuracy, as required.

Example 2.4. More examples from real-life applications; see Some disasters attributable to bad numerical computing.